# Structured summary: | Power Analysis for Longitudinal Clinical Trials with Run-In and Common Close Observations

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## Contents



*This document is a structured, section-by-section summary of the key points argued in the companion manuscript, "| Power Analysis for Longitudinal Clinical Trials with Run-In and Common Close Observations." It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.*

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# 1 Introduction

**Beyond its traditional roles, the run-in period has a distinct methodological value in rate-of-change trials: estimating individual baseline trajectories to raise power.**

- Conventional run-in uses include identifying compliant participants and washing out prior treatments.
- When the outcome is a rate of change, run-in observations estimate each participant's baseline trajectory.
- Incorporating those trajectories into the analysis model substantially increases statistical power.

**Frost and colleagues established the mixed-model foundation: run-in observations help most when slope variability is large relative to measurement error.**

- The key insight is that pre-randomization observations inform each participant's underlying trajectory.
- This reduces the residual variance of the treatment-effect estimator, and the gain grows with the slope-to-noise ratio.
- In Alzheimer's neuroimaging trials, where that ratio is favorable, they reported sample-size reductions of 30 to 45 percent.

**Wang and colleagues extended the framework with two-period models that jointly model run-in and post-randomization data rather than treating run-in as a covariate.**

- Joint modelling of both periods yields additional power gains of up to 15 percent.
- It permits unequal randomization ratios without power loss.
- Run-in observations contribute placebo-equivalent information for all participants.

**A parallel literature gives closed-form variance and sample-size formulae for the random-coefficient model without a run-in phase, which form the baseline this paper extends.**

- Lu and colleagues derived the cLDA estimator variance, its link to ANCOVA, and the conditions under which the two coincide.
- Hu and colleagues generalized these to random-coefficient models with monotone missing data and variable follow-up, illustrated with ADNI data.
- These results cover only the $J_0 = 0$ case and serve as the baseline extended here.

**The CRAN longpower package is the reference implementation of the baseline formulae, and our general result reduces to its routines at the appropriate boundaries.**

- longpower provides edland, diggle, and liu.liang routines for slope-comparison trials.
- The Stata package slopepower offers a complementary simulation-based route to the same class of calculations.

- Our formula reduces to the Ard-Edland slope variance at $J_0 = J_2 = 0$, and to the Diggle compound-symmetry form when also $\sigma_b^2 = 0$.
- The generalization thus embeds the existing longpower formulae as the baseline against which run-in and common-close gains are measured.

**No prior closed-form power formula handles an arbitrary number of run-in observations together with a separate common-close phase, the gap this paper fills.**

- The common-close phase observes all participants off-treatment after the randomized period.
- It supplies extra information about individual trajectories at no cost to the treatment comparison.
- Such post-treatment periods appear in delayed-start and randomized-withdrawal designs, but their GLS efficiency contribution has not been formally characterized.

**The paper presents three nested results, all derived from the GLS estimator via the Woodbury identity.**

- A closed-form variance for exactly two run-in observations (Section 3.2), then its generalization to arbitrary $J_0$ (Section 3.3).
- A further extension adding $J_2$ common-close observations (Section 3.4).
- All expressions are in terms of $\sigma^2$, $\sigma_b^2$, and spacing $t$, with numerical examples from Alzheimer's neuroimaging studies (Section 5).

## 1.1 General formulation

**The model accommodates a two-arm longitudinal trial whose observation schedule spans three distinct phases.**

- $N$ participants are observed at equally spaced time points.
- The phases are run-in (pre-randomization), treatment (post-randomization), and an optional common close (post-treatment).
- $Y_{ij}$ denotes the outcome for participant $i$ at time point $j$.

**The outcome model combines a random intercept and random slope with a three-component within-subject variance structure following Frost and colleagues.**

- $a_i$ and $b_i$ are the random intercept and slope, while $u_{ij}$ and $\varepsilon_{ij}$ are the between-visit and between-scan residuals.
- Both residual terms are independent across visits, so under a single scan per visit they enter every formula only through their sum.
- We therefore define the total within-visit residual $w_{ij} = u_{ij} + \varepsilon_{ij}$ with variance $\sigma^2$.

**The two within-visit components are identifiable only with replicate scans, so single-scan designs depend solely on their sum.**

- Separating $\sigma_u^2$ from $\sigma_\varepsilon^2$ requires a visit carrying replicate scans.

- For the single-scan designs treated here, only the sum $\sigma^2$ enters the treatment-effect variance.
- This reduction simplifies all subsequent variance expressions.

**Three fixed-effect slope parameters describe the run-in, placebo post-randomization, and treatment trajectories under equally spaced observations.**

- $\delta$ is the run-in rate of change common to all participants.
- $\beta$ is the placebo post-randomization rate of change, and $\gamma$ is the treatment effect on that rate.
- Observations are equally spaced with interval $t$, so $t_j = jt$.

**During the common close both arms revert to a shared off-treatment slope, modelled by deactivating the treatment indicator in that phase.**

- For $j > J_1$ all participants are off-treatment with slope $\beta + b_i$.
- This is implemented by setting $g_i = 0$ for every participant during the common close.
- Equivalently, the treatment indicator is replaced with $g_i \cdot I(1 \le j \le J_1)$ in the model.

## 1.2 Change-score parameterization

**Differencing relative to baseline removes the intercept and aligns the analysis with constrained longitudinal data analysis under compound symmetry.**

- The change score eliminates $\alpha$ and the random intercept $a_i$, leaving only two variance components.
- It is algebraically equivalent to cLDA, so the derived formulas apply to both analyses of run-in designs.
- By dropping $\sigma_a^2$ it avoids the anticonservative sample sizes that arise from ignoring intercept-slope correlation.

## 1.3 Covariance structure

**The shared baseline residual induces compound-symmetry correlation among change scores, so the residual covariance depends only on the summed within-visit variance.**

- The within-visit residuals are independent across visits with common variance $\sigma^2$.
- Differencing against the shared baseline $w_{i0}$ produces positive correlation, giving $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$.
- Both within-visit components enter $R$ identically, so only their sum is identified from single-scan designs.

## 1.4 GLS estimation and the Woodbury identity

**The estimand of interest is the treatment-effect variance, a single diagonal element of the inverted GLS information matrix.**

- The GLS estimator has covariance $(X'\Sigma^{-1}X)^{-1}$.
- $X$ is the fixed-effects design matrix and $\mathbf{c}$ is the stacked change-score vector.
- The target $\mathrm{Var}(\hat{\gamma})$ is the diagonal entry corresponding to $\gamma$.

### 1.4.1 Per-participant information, compact form

**The additive per-participant information form has two properties exploited later: a scalar rank-one correction and accommodation of heterogeneous designs.**

- The total information is the sum of per-participant contributions, with covariance given by its inverse.
- Under a single random slope, $\mathcal{J}_i$ is a scalar, making the correction a rank-one adjustment of the OLS-style information.
- Because contributions add, varying $J_0$, $J_2$, or dropout patterns are handled by summing over strata.

## 1.5 Case 1: Standard design, no run-in $(J_0 = 0,\ J_1 = 2,\ J_2 = 0)$

**The no-run-in case reproduces established results and serves as a verification baseline for the framework.**

- It reproduces Frost and colleagues, Section 3.1.1.
- It is the $J_0 = 0$ baseline of the random-coefficient power formulae of Lu and Hu and colleagues.
- It thereby provides a verification checkpoint for the present derivation.

**Without a run-in phase the design reduces to two post-randomization change scores and a two-parameter fixed-effects model.**

- There are $J = 2$ change scores at post-randomization times $t$ and $2t$.
- The absence of run-in removes the $\delta$ parameter from the model.
- The remaining fixed effects are the placebo slope $\beta$ and the treatment effect $\gamma$.

**The recovered variance matches Frost and supplies the per-group baseline for all run-in comparisons.**

- The expression agrees with Frost and colleagues, Section 3.1.1.
- For $m$ participants per group the variance scales as $(\sigma^2 + 2\sigma_b^2 t^2)/(mt^2)$.
- This quantity is the reference against which run-in designs are evaluated.

## 1.6 Case 2: Two run-in observations $(J_0 = 2,\ J_1 = 1,\ J_2 = 0)$

**Two run-in observations yield three change scores and a design matrix that distinguishes the run-in slope from the post-randomization slopes.**

- There are $J = 3$ change scores spanning the pre- and post-randomization times.
- The fixed-effects model now includes the run-in slope $\delta$ alongside $\beta$ and $\gamma$.

- The design matrix separates the run-in contribution from the post-randomization contribution.

**Because the random slope acts uniformly across visits, the random-effects design vector collapses to the scaled time distances from baseline.**

- The random slope $b_i$ applies to all time points alike.
- The random-effects design therefore reduces to $Z = t(-2, -1, 1)'$.
- The run-in change scores are differences of the pre-randomization outcomes from baseline.

**The treatment-effect variance is the relevant diagonal element of the inverted information matrix, obtained numerically and checked against the closed form.**

- $\mathrm{Var}(\hat{\gamma})$ is the $(3,3)$ element of $(X'\Sigma^{-1}X)^{-1}$.
- It is evaluated numerically in the accompanying R code.
- The numerical value is verified against the derived closed-form result.

**Completing the cofactor and determinant algebra yields the closed-form treatment-effect variance for the two-run-in design.**

- Define the auxiliary quantity $D = \sigma^2 + 5\sigma_b^2 t^2$.
- The $(3,3)$ cofactor simplifies to $6t^4/(aD)$.
- Dividing the cofactor by the determinant gives the boxed variance below.

**The run-in variance differs structurally from the Frost baseline by involving the residual variance in both numerator and denominator.**

- In the Frost formula $\sigma^2$ appears only additively in the numerator.
- Here $\sigma^2$ enters both the numerator and the denominator.
- This reflects the extra information extracted from pre-randomization observations.

## 1.7 General formula: $J_0$ run-in observations ($J_2 = 0$)

**The derivation now generalizes to an arbitrary number of run-in observations with no common close.**

- The design has arbitrary $J_0$ run-in and $J_1$ post-randomization observations.
- The common close is absent, so $J_2 = 0$.
- The total number of change scores is $J = J_0 + J_1$.

**The design matrix is built from run-in and post-randomization time vectors mapping to the three fixed-effect columns.**

- The columns correspond to the parameters $\delta$, $\beta$, and $\gamma$.
- $\mathbf{d}_{J_0}$ collects the negative run-in times scaled by $t$.
- $\mathbf{d}_{J_1}$ collects the positive post-randomization times scaled by $t$.

### 1.7.1 Computing $X'\Sigma^{-1}X$

**The Woodbury computation is organized around the sum-of-squares of the run-in and post-randomization time indices.**

- The derivation proceeds via the Woodbury identity.
- $S_{J_0}$ is the sum of squared run-in indices in closed form.
- $S_{J_1}$ is the corresponding sum over post-randomization indices.

**The structure of the information matrix reflects which parameters both groups inform and the opposing signs of the two phase time-sums.**

- Entries informed by both groups carry a factor of 2.
- The $\gamma$-related entries lack this factor because the placebo $\gamma$ column is zero.
- The positive $(\delta, \beta)$ cross-term arises from the opposite signs of the run-in and post-randomization time sums.

### 1.7.2 Numerical implementation

**For general run-in and post-randomization counts the variance has no compact closed form and must be evaluated as a complex ratio.**

- No simple closed-form expression exists for arbitrary $J_0$ and $J_1$.
- The variance depends on the sum-of-squares and sum quantities together with the dimension $J$.
- The Woodbury correction interacts with the information matrix to produce algebraically complex expressions.

**A matrix-based R implementation evaluates the exact variance for any configuration, with the closed-form cases as checkpoints.**

- The computation uses the matrix Woodbury identity in R code.
- It returns the exact variance for any $(J_0, J_1, J_2)$ configuration.
- The $J_0 = 0$ and $J_0 = 2$ closed forms serve as verification checkpoints.

### 1.7.3 Remark: recovery of the Edland and Diggle formulae

**Setting the run-in count to zero collapses the information matrix to a two-parameter block and yields the slope-comparison variance.**

- With $J_0 = 0$ the $\delta$ column vanishes, leaving a $2 \times 2$ block on $(\beta, \gamma)$.
- Sherman-Morrison cancellations collapse the Woodbury correction against the remaining structure.
- The result is the per-arm variance of the treatment-effect estimator for $n$ participants.

**The boundary case recovers the Edland, Diggle, and Frost formulae, confirming that the general result embeds the longpower routines as strict special cases.**

- The expression is the Ard-Edland slope-comparison formula implemented in the longpower edland routine.
- Setting the slope variance to zero gives compound symmetry and recovers the Diggle formula.
- The Frost formula is the two-visit specialization, so the general result quantifies gains accruing only from run-in or common-close observations.

### 1.8 General formula with common close ($J_2 \geq 0$)

**The final extension adds common-close observations after the treatment period to the change-score vector.**

- $J_2$ common-close observations follow the treatment period.
- All participants are off-treatment during this phase.
- The total dimension becomes $J = J_0 + J_1 + J_2$.

**In the common close the treatment effect persists only as an accumulated offset, fixing the design-matrix entry at the treatment-period value.**

- $\gamma$ enters only through the effect accumulated during the treatment period.
- It does not contribute to the slope during the common close.
- The $\gamma$ column therefore equals $J_1 t$ for all common-close observations in the treatment group.

**The slope and treatment columns of the design matrix encode the post-randomization times and the capped treatment exposure respectively.**

- The $\beta$ column holds the post-randomization times of the placebo slope across all later visits.
- The $\gamma$ column rises through the treatment period then stays at $J_1$ during the common close.
- The $\gamma$ column is identically zero for placebo participants.

**The common-close extension reuses the same Woodbury computation with only three quantities altered.**

- The dimension $J$ increases to include the common-close visits.
- The $X$ matrix changes only in its $\gamma$ column.
- The elements of the random-effects vector $Z$ are extended accordingly.

## 2 Power and Sample Size Formulas

**A single run-in observation never worsens precision, but the extra slope parameter it introduces can absorb nearly all of the information it contributes.**

- Adding one run-in observation introduces the parameter $\delta$, enlarging the fixed-effect dimension from 2 to 3.

- Across a wide numerical grid, $\mathrm{Var}(\hat{\gamma})|_{J_0=1}$ never exceeds $\mathrm{Var}(\hat{\gamma})|_{J_0=0}$.
- At some parameter values the gain is negligible, so the benefit of a single run-in observation should be quantified before extending trial duration.

### 2.1 Verification of closed-form results

**The closed-form expressions are validated against the matrix computation in two representative design configurations.**

- The verification confirms agreement between the analytic and matrix-based variances.
- The Frost standard design uses the two-parameter model with two post-baseline visits.
- The two-run-in case exercises the three-parameter model.

### 2.2 Validation of the general routine

**The Woodbury routine is validated against a direct GLS computation and a seeded Monte Carlo benchmark that both exercise the common-close path.**

- A direct computation assembles and inverts $\Sigma$ without the Woodbury identity over a grid of designs including $J_2 > 0$.
- Agreement with the Woodbury routine is at the level of numerical round-off.
- A seeded Monte Carlo at $(J_0, J_1, J_2) = (2, 2, 2)$ confirms that the empirical variance of $\hat{\gamma}$ matches the analytic value.

### 2.3 Application: Alzheimer's disease imaging trial

**The Alzheimer's application uses MIRIAD-derived parameters with the correctly combined within-visit residual, which is essential to reproduce Frost's reported sample sizes.**

- The slope variance and combined within-visit residual are taken from Frost's MIRIAD estimates.
- The example assumes a one-year observation interval and a treatment effect of one quarter.
- Using only the small measurement-error term understates the residual by two orders of magnitude and yields degenerate sample sizes.

## 3 Discussion

**The work formalizes and extends Frost's framework with explicit variance formulas for run-in and common-close designs.**

- The results provide explicit treatment-effect variance formulas.
- They cover designs incorporating both run-in and common-close observations.
- Three substantive points warrant further discussion.

**The benefit of run-in observations is governed by the slope-to-noise ratio and is largest for outcomes such as neuroimaging.**

- The efficiency gain depends on the ratio of slope variance to residual variance.
- When that ratio is large, even one or two run-in observations cut sample sizes substantially.
- For cognitive outcomes with a less favorable ratio, gains are modest and must be weighed against longer study duration.

**Common-close observations sharpen the treatment-effect estimator at low cost, with diminishing returns beyond the first few visits.**

- They inform individual trajectories without keeping participants on treatment.
- Both groups contribute to slope-variability estimation once treatment has ceased.
- Marginal benefit diminishes, but the first one or two visits give meaningful reductions when the slope-to-noise ratio is large.

**The derived formulas are consistent with Wang's two-period model while making the three phases explicit and separately optimizable.**

- Wang's common-slope scenario corresponds to setting the run-in and placebo slopes equal.
- Their different-slope scenario corresponds to the general case treated here.
- The present formulation separates run-in, treatment, and common-close phases, enabling per-phase optimization.

**The derivations rest on simplifying assumptions whose relaxation is feasible but typically forfeits closed-form simplicity.**

- They assume equally spaced observations, linear trajectories, and a random intercept-plus-slope structure.
- Extensions to unequal spacing, nonlinear progression, or richer random effects remain within the Woodbury framework but lose simple closed forms.
- No dropout is assumed, though the Dawson-Lagakos pattern-mixture approach can adjust for anticipated attrition.

# 4 Data availability statement

**All materials supporting the paper are openly available in the project repository.**

- These include the R package, the four companion manuscripts, and the Mathematica derivation scripts.
- The shared bibliography is likewise available.
- Specific paths are enumerated in the Reproducibility section.

# 5 Reproducibility

**The manuscript build relies on the in-package source together with a small set of testing and rendering packages.**

- Rendering used R version 4.4.x.
- The principal dependencies are the in-package source plus tinytest, rmarkdown, and bookdown.
- Supporting Mathematica symbolic derivations reside in the analysis scripts directory.

**The variance results are deterministic, and the Monte Carlo benchmark reported in the validation subsection uses a fixed seed.**

- Given fixed input parameters the formulas exercise no random-number generation.
- The validation subsection compares the Woodbury routine with a direct GLS computation over a grid that includes common-close configurations.
- Its Monte Carlo benchmark at $(J_0, J_1, J_2) = (2, 2, 2)$ sets the seed 20260418 at its entry point.

**The repository organizes the package source, derivation scripts, companion manuscripts, and bibliography into well-defined locations.**

- The R package source and the Mathematica scripts each have a dedicated directory.
- The three companion manuscripts share a report subtree.
- This manuscript source and the shared bibliography sit in the run-in power report directory.